

Research Assignment 2

Business Analytics Terminology

1 Business Analytics - Is the general of utilizing data, statistical analysis and iterative testing to make business decisions and solve corporate problems. It ties raw data to corporate strategy by focusing on the operational impact and business goals over technical analysis.

Example: Ecommerce brand reviews their historical purchase trends to understand which international market they should expand their shipping capabilities to next. - Harvard Business Review (hbr.org)

2 Data Analytic - Is the broader technical skillset that includes cleaning, transforming, modeling and inspecting raw data to discover patterns and draw out conclusions. While business analytics is purely corporate orientated, data analytics also includes a wide scientific domain that incorporates engineering and scientific data analysis.

Example: A software Engineer runs a script to clean thousands of ~~corrupt~~ ~~clean~~ of corrupted ~~has~~ database entries and group the valid user inputs by time zone - Ref: Evans, J.R (Business Analytics)

3 Business Intelligent (BI) - Is the discipline that involves aggregating, querying, and reporting historical data to describe the past, current state of business. Unlike advanced analytics, BI relies on descriptive data and dashboard to show managers exactly what is happening in the company right now.

Example: Regional sales directors look at a live PowerBI dashboard every morning to see if their team met yesterday's revenue target.
References - McKinsey & Company insights

4 Data-Driven Decision Making - Is a Corporate management philosophy that is driven on data being validated as a metric to support business decision instead of intuition or executive guesswork. It encourages organization culture that demands facts before greenlighting expensive long term projects

Examples

A streaming service decides to ~~renew~~ renew a television series entirely because the user completion rates crossed a pre-established percentage threshold - Ref Harvard Business Review (hbr.org)

5 Big Data - This is the term for massive, complex sets of data that cannot be efficiently processed or analyzed using traditional relational database software. It is typically characterized by the "3 Vs": high volume, high velocity of real-time generation and wide variety of format

Example

A ridesharing company continuously processes billions of live GPS coordinates, traffic updates, and user request to calculate dynamic surge pricing. Ref Gartner IT Glossary / Investopedia

6 Descriptive Analytics - Is the initial processing stage that answers the fundamental question, "What happened?". It summarizes historical data with charts, tables and report to give teams a clear, retroactive view of past performances

Example

A retail chain publishes an annual financial report that shows that holiday sales rose by 12% compared to the previous year. Evans JR. (Business Analytics)

7. Diagnostic Analytics - Is the form advanced analytics takes the historical data a step further to answer the core question, "Why did it happen?" It uses drills-down, data discovery and correlations to isolate the root causes of unexpected anomalies or trends

Real-world Example

An application provider investigates a sudden spike in user churn and finds it was caused by a software bug introduced in a recent update
McKinsey & Company Insights

8. Predictive Analytics - Is the practice uses historical data patterns machine learning algorithms and statistical modeling to answer the question "What is likely to happen?". It builds forecasts to estimate future trends to help companies prepare for upcoming market shifts

Real World Example

A credit card company evaluates a customer's spending history and repayment habits to calculate the statistical likelihood of them defaulting on a loan - Investopedia

9. Prescriptive Analytics - Is the highest tier of analytics which answers "What should we do about it?". It uses optimization techniques and simulation algorithms to automatically recommend the most effective course of action to achieve a specific corporate goal

Real-World Example

An airline uses automated software to constantly adjust seat prices based on fluctuating demand, competitor pricing, historical booking velocity

10. Data Literacy - Is a modern professional Competency that defined as the ability to read, work with, analyze and communicate data effectively in a work setting. It ensures that non-technical employees can critically evaluate metric and contribute to data-driven discussion.

Real World Example

A human resources manager successfully interprets an internal employee turnover chart to build a better staff retention program [Harvard Business Review] hbr.org, Data, measurement & Reporting

11 Structured Data - This refers to highly structured, standardized information that conforms to rigid predefined data model that makes it easily searchable in relational database. It is typical in rows and columns such as SQL table or Excel sheets - where each field has a clearly defined data type (date, integers or strings)

Real World Example

An airline database houses customer booking records, where every row contains distinct columns for the passenger ID, flight number, departure date, and seat assignment - Investopedia

12 Unstructured Data - This is information that does not possess conceptual structure or predefined data model, making it complex to compile, search and analyse using traditional relational tool. It constitutes the vast majority of real world corporate data that include assets such as native text documents, email threads, audio recording, video clips,

Real world example

A customer service department archives thousands of recorded call phone conversations and scanned handwritten compliant letter received over the past calendar year. Mckinsey & Company insights

13 Data Warehouse - This is a centralized corporate computing repository designed to aggregate, clean, and store vast volumes of historical data extracted from multiple disparate operating systems across an organization. It acts as single source of truth optimized specifically for analytical querying and reporting separate for live operational transaction database.

Real World example

A multinational retail conglomerate integrates point of sale data, supply chain logistics records and HR payroll files into single snowflake database for executive review (Evan, JR Business Analytics)

14 Data Mining - Is the analytical process of exploring massive datasets to automatically discover hidden, valid and previously unknown patterns correlations and anomalies. It combines statistical modeling, mathematical algorithms and machine learning techniques to turn raw corporate data into actionable business intelligences.

Real World Example

A credit card network runs anomaly-detection algorithms across billions of transactions to identify sudden, irregular buying spikes that are indicative of fraudulent behavior (Statista)

15 Data Visualization - Is the practice of translating quantitative data metrics and complex statistical findings into clear graphical representations, charts and maps. Its primary goal is to leverage human visual processing to help non-technical corporate decision-makers quickly digest complex trends, outliers and patterns.

eg A logistics firm design a dynamic heat map that uses shifting colour gradients to display regional shipping delay frequencies across global transit route (Harvard Business Review)

16 Dashboard - Is a live, interactive data visualization interface that ~~make~~ ^{aggregates} and displays critical performance metrics and real-time business health indicators on a single screen. It is linked directly to underlying data streams, allowing managers to monitor operational efficiency and drill down into granular data on demand.

Real World Example

A SaaS Company implement a cloud-base dashboard that monitors live platform server uptime, active user ~~counts~~ and concurrent customer help tickets - Google Analytics help & skillshop

17 Metric - This is any quantifiable, measurable value used to track monitor and assess the status of specific business activity or operational process. While highly useful for baseline tracking, a metrics lacks strategic prioritization, so it simply represents a neutral unit of data measurement.

Real World Example

A Corporate marketing ~~represents~~ ^{department counts} a number the exact number of total impressions or individual image views an uploaded corporate blog receives in a week (Evam JR Business Analytics)

18 Key Performance Indicator (KPI)

This is a highly strategic, selected that directly measures how effectively a company is achieving its core, high-priority business objectives. Every KPI is metric, but a metric only becomes a KPI if it directly influences critical strategic decisions and corporate survival.

While a total website visits is a baseline metric, an online banking firm's online loan application completion rate is a core corporate KPI
Harvard Business Review (hbr.org)

19 Benchmark - Is a standardized reference or baseline metric used to evaluate the performance, quality or efficiency of a company's operations.

Benchmark can be internal (comparing against historical company performance) or external (comparing against competitors or industry averages)

Real World Example

A boutique hotel chain measures its average occupancy rate against the regional luxury hospitality standard of 72% to gauge market competitiveness - Statista

20 Segmentation - Is the data-driven process of dividing a broad customer base or target market into distinct, manageable sub-groups based on shared characteristics. These groups are organized by characteristics such as demographics, location, historical purchasing behavior, psychographic tendencies

Real World Example

A fitness brand segments its primary email list into "casual gym-goers" and "competitive marathon runners" to deliver tailored product advertisement - HubSpot Academy

21 Conversion Rate - Is the total percentage of total website visitors or leads who successfully complete a specific, desired action out of the total pool of participants. It's a direct indicator of the user experience, design effectiveness and marketing message alignment

Real World Example

An online marketplace receives 10,000 unique landing page visitors over the weekend, resulting in 300 completed purchases for a 3% conversion rate - Google Analytics Help & Skillshop

22 Customer Acquisition Cost (CAC) - represent the total Comprehensive investment required to convince a potential consumer to purchase a company's product or service. It is calculated by dividing all the sales and marketing expenditures in a given timeframe by the total volume of new customers secured

Real World Example

A software brand spends R150000 across digital advertisements and sales salaries in Q1, successfully gaining 1000 new premium accounts for CAC of R150 per user - Hubspot Academy

23 Customer Life Value - Is a predictive metric that estimates the total net profit or gross income a business expects to generate from one single customer account throughout their entire ongoing relationship. When evaluated alongside CAC, the CLV:CAC ratio serves as a core indicator of corporate sustainability; health enterprise ideally target a ratio of 3:1 or higher

Real World Example

A gym member pay R500pm fee and stay registered for an average of 24 months before cancelling, resulting in baseline gross CLV of R12000 - Investopedia

24 Churn Rate - Is the percentage rate at which existing customer terminate their commercial relationship, cancel active subscriptions or stop buying from business over a designated period. A churn rate is a critical warning sign that indicates problems with the product, service or price to value alignment

eg A streaming network starts the month with 100000 active subscribers, but 4000 of those users cancel the account before month ends for a 4% monthly churn rate - McKinsey & Company insights

25 Customer Retention rate - Is the percentage of existing customers that a business successfully keeps active and engaged over a specific timeframe excluding newly acquired accounts. It is the direct mathematical inversion of a company's churn rate

Real World Example

A software company starts the year with 500 enterprise clients, loses 25 of them and signs 50 new ones; its strict retention rate for the original cohort is at 95% - Harvard Business Review

26 Net Promoter Score (NPS) - Is a standardized market research metric that measures customer loyalty, satisfaction and brand advocacy by asking single question "on a scale of 0 to 10, 'How likely are you to recommend us?' Survey respondents are categorized into promoters (9-10), Passives (7-8), and detractors (0-6). With the final score calculated by subtracting the detractor percentage from the promoter percentage

Real World Example

A mobile carrier surveys its user base, finding that 60% are promoters and 15% are detractors, resulting in an overall corporate NPS of 45% - Forrester & Company / Investopedia

27 Average Order Value - Is a metric average monetary amount spent by a customer every time they place an order through a company's sales channel. It is calculated by dividing total generated revenue by the total absolute of discrete customer transactions processed within that period

Real World Example

An online clothing boutique generates \$50,000 in daily revenue across 500 independent shopping cart check-outs for an AOV of \$100 per transaction
Google Analytics help and Skillshop

28 Sale funnels - Is a Conceptual marketing model that maps out the theoretical journey that consumer takes from the first moment of brand discovery ~~the~~ claim to the final conversion. It is traditionally broken into hierarchical stages - Awareness, interest, Consideration, intent and purchase to help analysts determine where prospects fall out of the buying journey

Real World Example

An automobile manufacturer tracks 50000 website visits which turn into dealership test drives, ultimately resulting in 200 finalised vehicle sales - Hubspot Academy

29 Marketing & Campaign Analytics - Is a strategic, coordinated series of promotional efforts that are designed to communicate a unified message and achieve a specific business goal over defined timeframe. These initiative leverages a variety of communication channels such as a paid search, television ads, events and email marketing to influence consumer behavior

Real World Example

A beverage company launches a multi-billion rand summer advertising push across social media platforms to introduce a new zero-sugar soft drink - Hubspot Academy

30 Post-Campaign Analysis - Is a comprehensive evaluation process conducted after a promotional initiative concludes to measure performance against its initial goal. It relies on data metrics such as incremental revenue and ROI to determine what worked, what failed, how future marketing budget should be allocated

eg A hardware retailer analyzes data from its seasonal promotional to figure out why video ads perform better than radio ads. - McKinsey & Company

31 Incremental Revenue - Is the additional, net new revenue generated as a result of a specific marketing or business intervention. It is isolated sales that would not have naturally occurred without the active campaign running, filtering out baseline sales trends.

Real World Example

A fashion brand typically pulls in R1000000 a week; during a flash sale email campaign, they generated R1400000 with campaign driving R400000 in incremental revenue - Evans. JR Business Analysts

32 Return On Investment (ROI) - Is a standard profitability metric used to evaluate the financial efficiency or performance of a business expenditure relative to its initial cost. It is calculated by subtracting the investment cost from the net profit gain, then dividing that result by the total initial investment cost, expressed as %

Real World Example

A firm invests R20000 in updating its retail storefront and tracks a subsequent R30000 increase in net profits, yielding an investment ROI of 50% - Investopedia

33 Return on Ad Spend (ROAS) - Is a focused digital marketing metric that tracks the gross revenue generated for every single currency unit spent specifically on running advertisements. Unlike ROI, which evaluates net profitability after all business expenses, ROAS evaluate top line gross marketing performance to measure ad campaign efficiency

eg An ecommerce company spend R5000 on google search advertisements and generate R25000 in direct product sales, resulting in a 5:1 ROAS
- Google Analytics Help & Skillshop

34 Click-through Rate (CTR) - Is the total percentage viewer who clicked on a specific hyperlink, digital advertisement or email element out of the total number of individual who viewed it. It act as a direct measure of an advertisement's visual appeal, keyword relevance and copy effectiveness.

Real World Example

A sponsored banner ad is displayed on a news website 10,000 times (impressions) and receives 200 clicks, resulting in a click-through rate of two percent - Google Analytics help & Skillshop

35 Bounce Rate - Is the percentage of website visitors who leave a platform after viewing only one single page without triggering any other actions or navigating deeper into the site. A high bounce rate typically signals technical landing page problems, slowing loading times or irrelevant content

Real World Example

A user clicks an online to shoe store, looks at the landing page for 3 seconds without clicking anything and hits back button - Google Analytics help and Skillshop

36 A/B Testing - Is a controlled experimental practice where two distinct version of a digital asset (version A and version B) are shown to separate user segments simultaneously. By altering a single variable such as headline colour or button placement, analysts can determine which version has statistically higher conversion rate

Real World Example

A tech company shows a blue "Sign up" button to half of its web visitors and green button to the other half to see which colour drives more registration - Harvard Business Review (hbr.org)

37 Lead Generation - Is the strategic process of identifying, attracting and converting strangers or prospects into individual who have actively demonstrated interest in company's products or service. It focuses on collecting consumer data such as contact, emails, phone numbers

Real World example

A commercial company accounting agency offer a free downloadable tax guide in exchange for a business owner's name, phone number and company email address - Hubspot Academy

38 Marketing Attribution - Is an analytical framework used to evaluate and assign ~~ment~~ financial credit to the different marketing touchpoint a consumer interacts with on their path to purchase. It help business understand which specific combination of channel (eg social ads, organic search, direct email)

Real World Example

A buyer clicks a social media ad, reads a follow-up newsletter and searches the brand on google before buying; attribution models determine how much credit each channel get - Google Analytics Help & Skillshop